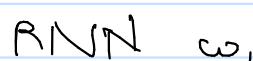


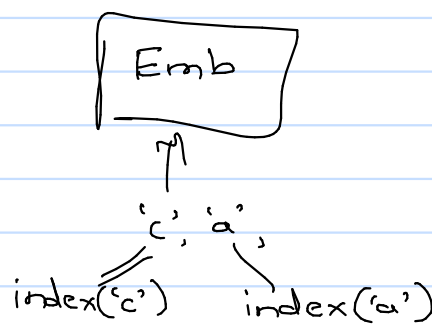
$$\begin{array}{c} \overbrace{\begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array}}^m \\ \underbrace{\begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array}}_n \quad \underbrace{\begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array}}_k \\ \frac{\text{sum}_1}{n} \quad \frac{\text{sum}_2}{k} \end{array}$$


$$\begin{aligned} o_1 &= f(\underline{w}_1, 'h', h_0) \\ o_2 &= f(\underline{w}_1, 'e', h_1) \\ o_3 &= f(\underline{w}_1, 'l', h_2) \\ o_4 &= f(\underline{w}_1, 'l', h_3) \end{aligned}$$

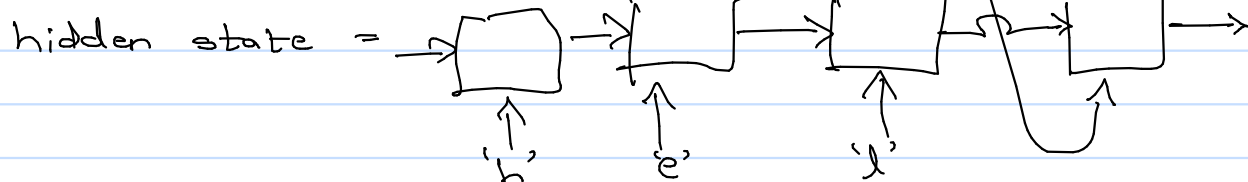
$$\text{loss} = \left(\begin{array}{l} \text{cel}(o_1, 'e') \\ + \text{cel}(o_2, 'l') \\ + \text{cel}(o_3, 'l') \\ + \underbrace{\text{cel}(o_4, 'o')} \end{array} \right)$$

$$\begin{aligned} h_1 &= f'(\omega_1, h_0, 'h') \\ h_2 &= f'(\omega_1, h_1, 'e') \\ h_3 &= f'(\omega_1, h_2, 'l') \end{aligned}$$

$$\begin{array}{ccccccc}
 c_1 & c_2 & c_3 & \dots & c_{199} & \rightarrow \text{inp} \\
 \hline
 c_2 & c_3 & c_4 & \dots & c_{200} & \rightarrow \text{target}
 \end{array}$$



hidden state



hello
hell ← inp
ello ← target

$$\left(\frac{\partial u_x}{\partial x}\right)^2 + u_x \frac{\partial^2 u_x}{\partial x^2} + \frac{\partial u_y}{\partial x} \frac{\partial u_z}{\partial y} + u_y \left(\frac{\partial^2 u_x}{\partial x \partial y}\right)$$